

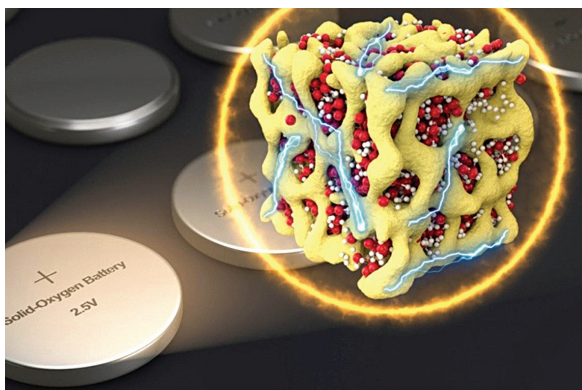


Fig. 2: Nanolithia particles of lithium-oxides shown in yellow in red cobalt-oxide lattice

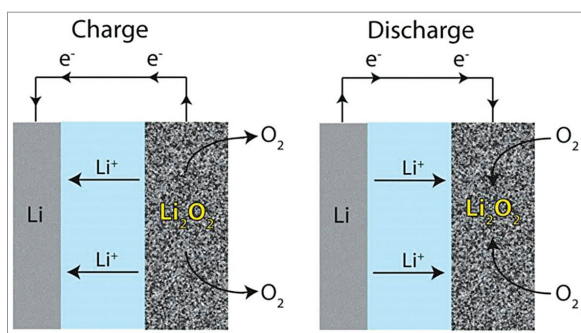


Fig. 3: Charging and discharging mechanism of lithium-air battery

very bad electron conductor. It clogs the cathode pores and reduces the reversible reaction. The problem can be solved by adding lithium iodide as electrolyte additive to help break up the solid discharge product to form lithium hydroxide. As lithium hydroxide is soluble, it would not block the active carbon surface that generates the supply voltage.

2. Organic polymer electrolyte is used, which is mechanically stable as well as a good ionic conductor. The mechanical hardness of the polymer segment inhibits electrical shorting due to dendrite formation.

3. Researchers sometimes use redox mediators—molecules that go into a solution and help the reaction process. It brings down the difference between the charge and discharge to its lowest value to date of only 0.2V. So, battery can be made with 90% efficiency.

4. Protective layers are used on the anode so that lithium metal does not react with other components of air.

5. Sometimes the lithium-air battery is simply charged and discharged for a few cycles in a pure CO_2 atmosphere when a dense mesh of lithium-carbonate crystals is formed on the surface of lithium anode, which acts as a filter and prevents the migration of water (H_2O), oxygen (O_2), CO_2 , and nitrogen to lithium anode, but allows lithium ions to move in the electrolyte selectively. So, unwanted side reactions are prevented.

6. By using special cathode made from molybdenum-di-sulphide, a thin film of lithium-peroxide is formed on cathode surface during normal operation,

which is inert and protects the cathode from unwanted reaction. In this way batteries can be recharged 750 times.

7. In 2018, a new concept envisaged embedding nano compounds of lithium-oxides in a sponge-like lattice of cobalt-oxide, called nanolithia, as shown in Fig. 2. The nanolithia particles of lithium and oxygen are generally very unstable. But these particles become stable when housed in a cobalt-oxide matrix and, surprisingly, the aggregate acts as a fine catalyst to speed up the chemical reaction. In this variant the same kind of electrochemical reaction takes place between lithium and oxygen during charging and discharging but without letting the oxygen revert to its gaseous form. The bound oxygen in nanolithia does the job. In lithium-air batteries the voltage difference between charging and discharging is 1.3V, which reduces the charging efficiency. Use of nanolithia reduces the voltage loss from 1.3V

to 0.24V. So, only 8% of electrical energy is turned to heat, resulting in the upgradation of charging efficiency to about 90%.

Originally proposed in the 1970s as a possible power source for battery electric vehicles, the risk-to-benefit ratio was perceived as too high to pursue the new technology. Nevertheless, due to absence of other alternatives to high specific energy rechargeable batteries, and due to some initial promising results from laboratories, lithium-air batteries recaptured scientific interest late in the first decade of the 2000s and also due to advances in material sciences.

Electrochemistry

A lithium-air cell creates voltage from the availability of oxygen molecules at the positive electrode. Oxygen reacts with the positively charged lithium ions to form lithium-peroxide (Li_2O_2) and generates electricity. Electrons are drawn out of the electrode and such a battery is empty (discharged) if no more Li_2O_2 can be formed. The battery is fully discharged when all the pores of the cathode are filled up with Li_2O_2 .

In general, lithium ions move between the anode and the cathode across the electrolyte. Under discharge, electrons follow the external circuit and do electrical work and lithium ions migrate to the cathode. During charging the lithium metal is electroplated onto the anode, freeing oxygen at the cathode (see Fig. 3).

Reaction during discharging is as follows:

Anode reaction: $\text{Li} = \text{Li-ion} (\text{Li}^+) + \text{electron} (\text{e}^-)$

Cathode reaction: $\text{O}_2 + \text{Li}^+ + \text{e}^- = \text{LiO}_2$
 $2\text{LiO}_2 = \text{Li}_2\text{O}_2 + \text{O}_2$

$\text{LiO}_2 + \text{Li}^+ + \text{e}^- = \text{Li}_2\text{O}_2$ ($E_0 = 3.1\text{V}$) where E_0 is standard cell potential.

Reaction during charging is as follows:

Anode reaction: $\text{Li-ion} + \text{electron} = \text{Li metal (plates onto anode)}$

Cathode reaction: $\text{Li}_2\text{O}_2 = \text{Li-ion}$